

Solutions to JEE Advanced - 3 | JEE 2022 | Paper - 2

PHYSICS

CHEMISTRY

1.(2) (1) Given that for Y : $t_{75\%} = 1.5t_{50\%}$, thus order w.r.t. "Y" is zero

(2) $t_{50\%}$ for Z becomes half if initial concentration of Z is doubled thus order w.r.t. Z is 2 $\left[t_{50\%} = \frac{1}{K_a} \right]$

(3) Rate of the reaction becomes 4 times when concentration of both "X" and "Z" are doubled. As order w.r.t. Z is 2 thus it must be zero w.r.t. X

Overall order of the reaction = $0 + 0 + 2 = 2$

2.(0) ${}^{238}_{92}\text{U} \longrightarrow {}^{218}_{84}\text{Po} + 5 {}^4_2\alpha + 2 {}^0_{-1}\beta$ [x = 2]

${}^{218}_{84}\text{Po} \longrightarrow {}^{214}_{83}\text{Bi} + {}^4_2\alpha + {}^0_{-1}\beta$ [y = 1]

${}^{214}_{83}\text{Bi} \longrightarrow {}^{206}_{82}\text{Pb} + 2 {}^4_2\alpha + 3 {}^0_{-1}\beta$ [z = 3]

$(x + y - z) = 2 + 1 - 3 = 0$

3.(3) Compound "X" must have some magnetic nature [Para/Ferro/Ferri-magnetic]

MnO , Fe_2O_3 , NiO are Antiferromagnetic

NaCl and $\text{ZnS} \rightarrow$ Diamagnetic

Fe_3O_4 is Ferrimagnetic

CrO_2 , Mn are Ferromagnetic

4.(4)

S. No.	System	Axial Lengths	Axial Angles
1.	Cubic	$a = b = c$	$\alpha = \beta = \gamma = 90^\circ$
2.	Tetragonal	$a = b \neq c$	$\alpha = \beta = \gamma = 90^\circ$
3.	Orthorhombic	$a \neq b \neq c$	$\alpha = \beta = \gamma = 90^\circ$
4.	Hexagonal	$a = b \neq c$	$\alpha = \beta = \gamma \neq 90^\circ$
5.	Rhombohedral	$a = b \neq c$	$\alpha = \beta = 90^\circ, \gamma = 120^\circ$
6.	Monoclinic	$a \neq b \neq c$	$\alpha = \gamma = 90^\circ, \beta \neq 90^\circ$
7.	Triclinic	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma \neq 90^\circ$

These four belongs to the mentioned category

5.(3) To create 1 Schottky defect per unit cell of NaCl we have to remove atleast 1Na^+ ion from the body center location and 2Cl^- ions from the face center locations.

Thus $x = 1$ and $y = 2 \Rightarrow (x + y) = 3$

6.(4) HA acid

$\text{pH} = 1.70$

$[\text{H}^+] = 2 \times 10^{-2} \text{M} = \alpha C$

$C = 10^{-1} \text{M}$

$\Rightarrow \alpha = 0.2$

$$\frac{\pi_{\text{HA}}}{\pi_{\text{HB}}} = \frac{(1 + \alpha)\text{CRT}}{(1 + \alpha')C'\text{RT}} = \frac{(1 + 0.2)10^{-1}}{(1 + 0.5)2 \times 10^{-1}} = 0.4$$

HB acid

$\text{pH} = 1$

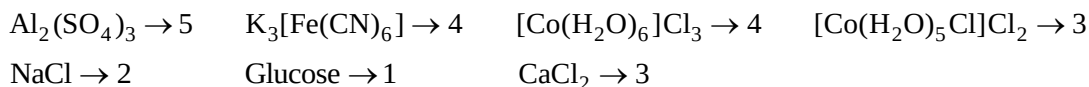
$[\text{H}^+] = 10^{-1} \text{M} = \alpha' C'$

$C' = 0.2 \text{M}$

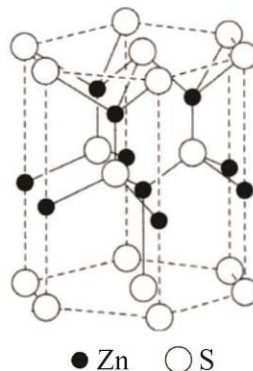
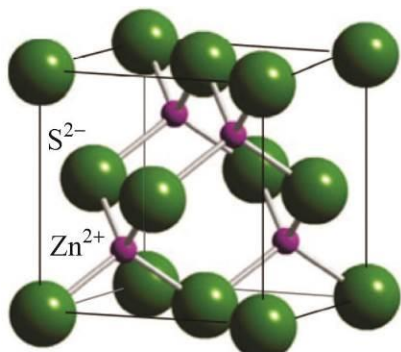
$\Rightarrow \alpha' = 0.5$

7.(BCD) Osmotic pressure and boiling point increases with the Vant Hoff factor while vapour pressure and freezing point decreases with Vant Hoff factor.

Vant Hoff factors.



8.(BD)



(Zinc blende)

(Wurtzite)

- | | | | |
|----|--|----------|----------|
| 1. | C.N. of Zn^{2+} | 4 | 4 |
| 2. | Zn^{2+} shared by adjacent unit cells | ✗ | ✓ |
| 3. | ZnS formula units | 4 | 6 |
| 4. | 50% of total voids occupied | NOT true | NOT true |

9.(ABC) $\Delta H_{\text{mix}} < 0$, $\Delta S_{\text{mix}} > 0$, $\Delta V_{\text{mix}} < 0$ are properties of a non-ideal solution with negative deviation from the Raoult's law. (A), (B) and (C) are the non-ideal solutions with negative deviation.

10.(ACD) According to collision theory

$$K_{\text{obs}} = PK_{\text{theo}}$$

$$K_{\text{obs}} = P[A_{\text{theo}}]e^{-E_A/RT}$$

$$A_{\text{obs}} = PA_{\text{theo}}$$

For $P > 1$

$$A_{\text{obs}} > A_{\text{theo}}$$

Also $K_{\text{theo}} = Ze^{-E_A/RT}$

Thus rate of a reaction

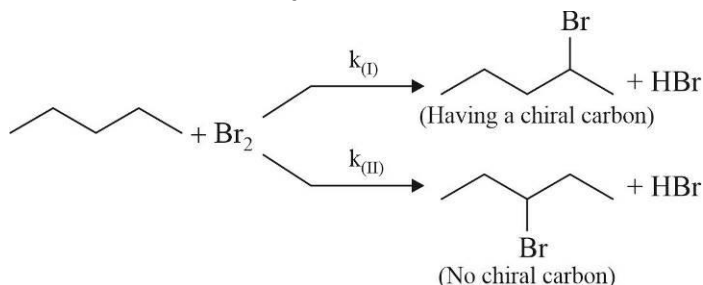
- Increases with Z (collision frequency)
- Depends upon temperature (T)
- Increases with the Boltzmann factor $[e^{-E_A/RT}]$

11.(ABCD)

- (A) Emission of an alpha particle increases $\frac{N}{P}$ ratio, thus such nuclei then undergo β -emission to reduce that ratio
- (B) K-capture gives isobar of the parent nuclei
- (C) Neutron emission gives an isotope of the parent nuclei
- (D) Positron emission is used by a nuclei with low $\frac{N}{P}$ ratio or high $\frac{P}{N}$ ratio

- 12.(BCD) Except (A) all other statements are correct. In a 3D HCP unit cell each sphere is contributing some of its volume in adjacent unit cells.

13.(80)



$$\% \text{ of product with chiral carbon} = \frac{k_{\text{(I)}}}{k_{\text{(I)}} + k_{\text{(II)}}} \times 100 = \frac{8 \times 10^{-2}}{8 \times 10^{-2} + 2 \times 10^{-2}} = 80\%$$

- 14.(74) A truncated octahedron has 14 faces, 36 edges and 24 vertices. Number of faces (F), vertices (V) and edges (E) are related by formula, $F + V = E + 2$

- 15.(117.52) Vapour pressure of the resulting solution can be determine using $\frac{P^\circ - P_s}{P_s} = \frac{m M_{\text{solvent}}^\circ}{1000}$

m = Molality = 2 molal

While using fractional distillation

$$Y_B = \frac{P_B^\circ X_B}{P_B^\circ X_B + P_A^\circ X_A} = \frac{150 \times 4}{150 \times 4 + 2 \times 100} = \frac{600}{800} = \frac{3}{4} \Rightarrow Y_A = \frac{1}{4}$$

$$M_{\text{solvent}}^\circ = \frac{3}{4}[80] + \frac{1}{4}[100] = 85$$

[1st distillate]

$$P_{[1^{\text{st}} \text{ distillate}]}^\circ = \frac{1}{4}[100] + \frac{3}{4}[150] = 137.5$$

$$\frac{137.5 - P_s}{P_s} = \frac{2[85]}{1000} \Rightarrow 137.5 - P_s = 0.17 P_s \Rightarrow P_s = \frac{137.5}{1.17} = 117.5 \text{ torr}$$

- 16.(1.40)
- $$\begin{array}{ccccccc} \text{A}^{1+} \text{B}^{2-} & \xrightarrow{\text{(i)}} & \text{A}^{1+} \text{B}^{2-} & \xrightarrow{\text{(ii)}} & \text{A}^{1+} \text{B}^{2-} \text{C}^{2+} & \xrightarrow{\text{(iii)}} & \text{A}^{1+} \text{B}^{2-} \text{C}^{2+} & \xrightarrow{\text{(iv)}} & \text{A}^{1+} \text{B}^{2-} \text{C}^{2+} \text{D}^{1-} \\ \left[\begin{array}{l} \text{Cations} = 8 \\ \text{Anions} = 4 \end{array} \right] & & \left[\begin{array}{l} \text{Cations} = 8 \\ \text{Anions} = 1 \end{array} \right] & & \left[\begin{array}{l} \text{Cations} = 8+3=11 \\ \text{Anions} = 1 \end{array} \right] & & \left[\begin{array}{l} \text{Cations} = 11-4=7 \\ \text{Anions} = 1 \end{array} \right] & & \left[\begin{array}{l} \text{Cations} = 7 \\ \text{Anions} = 5 \end{array} \right] \end{array}$$

In final compound "X"

$$\frac{\text{Number of cations}}{\text{Number of anions}} = \frac{7}{5} = 1.4$$

- 17.(17.24) Osmotic pressure of solution A = $\pi_A = i_A C_A RT = 5[0.2]RT = RT$

$$\text{Osmotic pressure of solution B} = \pi_B = i_B C_B RT = 3[0.1]RT = 0.3RT$$

$\therefore$ Solution A is hypertonic to solution B and water will be flow from B to A.

To prevent that flow pressure "P" pressure must be applied over surface of A

$$\text{Such that } P = \pi_A - \pi_B = 0.7RT = 0.7[0.0821][300] = 17.24 \text{ atm}$$

- 18.(2880) As $\log_{10} k = \log_{10} A - \frac{E_A}{2.303R} \left[\frac{1}{T} \right]$ and $\log \frac{k_{T_2}}{k_{T_1}} = \frac{E_A}{2.303R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right]$

$$0.60 = \frac{E_A}{2.303R} \left[\frac{1}{300} - \frac{1}{320} \right]; \quad 0.60 = (\text{slope}) \left[\frac{20}{300 \times 320} \right] \quad (\text{Slope}) = 0.60 \times 300 \times 16 = 2880$$

MATHEMATICS

1.(3) Equate LHS and RHS and match the coefficients.

2.(1) Doubt at $x = 0$ and 1

Apply *LHD, RHD*

At $x = 0$ $LHD \neq RHD$

3.(7) $x^3 + 3x + 6 = t$

$$3(x^2 + 1)dx = dt$$

$$(x^2 + 1)dx = \frac{dt}{3}$$

$$I = \frac{1}{3} \int \frac{dt}{t^{1/3}} = \frac{1}{3} \times \frac{t^{2/3}}{\frac{2}{3}} + c = \frac{1}{2} \cdot t^{2/3} + c$$

4.(1) Divide by x^2

$$\int \frac{1 - \frac{1}{x^2}}{\left(x^2 + \frac{1}{x^2} + 3\right) \tan^{-1}\left(x + \frac{1}{x}\right)} dx$$

$$x + \frac{1}{x} = t$$

$$= \int \frac{dt}{(t^2 + 1) \cdot \tan^{-1} t} = \log \tan^{-1} t + c$$

$$K = 1$$

5.(0) $\int \frac{\sin x}{\cos^2 x} dx + \int \frac{\cos x}{\sin^2 x} dx$

Putting $\cos x = t$ and $\sin x = t$ respectively

$$\Rightarrow \frac{-(\cos x)^{-2+1}}{-2+1} + \frac{(\sin x)^{-2+1}}{-2+1} + c$$

$$\Rightarrow \sec x - \csc x + c \Rightarrow K_1 = 1, K_2 = -1$$

6.(2) Differentiating $y^3 - x^2y + 5y - 2x = 0$

w.r.t x , we get

$$3y^2y' - 2xy' - x^2y' + 5y' - 2 = 0 \text{ or } y' = \frac{2xy + 2}{3y^2 - x^2 + 5}$$

$$\text{or } y'(0, 0) = \frac{2}{5} \text{ differentiating}$$

$$x^4 - x^3y^2 + 5x + 2y = 0$$

w.r.t. x , we get

$$4x^3 - 3x^2y^2 - 2x^3yy' + 5 + 2y' = 0 \text{ or } y' = \frac{3x^2y^2 - 4x^3 - 5}{2 - 2x^3y} \text{ or } y'(0, 0) = -\frac{5}{2}$$

Thus, both the curves intersect at right angle

$$7.(AD) \quad I = \frac{\sec^2 x dx}{(\sec x + \tan x)^{9/2}}$$

$$I = \frac{\sec x \sec x (\sec x + \tan x) dx}{(\sec x + \tan x)^{11/2}} \dots\dots\dots(1)$$

$$\sec x + \tan x = t$$

$$(\sec x \tan x + \sec^2 x) dx = dt$$

$$\sec x (\sec x + \tan x) dx = dt \quad ; \quad \sec x - \tan x = \frac{1}{t} \quad ; \quad 2 \sec x = t + \frac{1}{t}$$

Now put these value of t in equation (1)

$$I = \frac{1}{2} \int \frac{\left(t + \frac{1}{t}\right) dt}{t^{11/2}}$$

$$I = \frac{1}{2} \int (t^{-3/2} + t^{-13/2}) dt \quad ; \quad = \frac{1}{2} \int \left(\frac{t^{-7/2}}{-7/2} + \frac{+t^{-11/2}}{-11/2} \right) + c$$

$$= - \left[\frac{1}{7t^{7/2}} + \frac{1}{11t^{11/2}} \right] + c = - \frac{1}{t^{11/2}} \left[\frac{1}{11} + \frac{1}{7} t^2 \right] + c = - \frac{1}{(\sec x + \tan x)^{11/2}} \left[\frac{1}{11} + \frac{1}{7} (\sec x + \tan x)^2 \right] + c$$

$$8.(AD) \quad f'(x) = \frac{1}{(1+x)^2} > 0, \forall x \in [0, \infty) \text{ and range } [0, 1)$$

Function is one-one but not onto

$$9.(AD) \quad \text{We have } f(x) = (x-1)(x-2)(x-3) \text{ and } f(1) = f(2) = f(3) = 0 \text{ is not one-one}$$

For each $x, y \in R$, there exists $x \in R$ such that $f(x) = y$.

Therefore f is onto.

Hence $f : R \rightarrow R$ is onto but not one-one

$$10.(BD) \quad \text{LHL} \quad x = -h$$

$$\lim_{h \rightarrow 0} (-\sin h + \cos h) \frac{-1}{\sin h}$$

$$= e^{h \rightarrow 0} (-\sin h + \cos h - 1) \cdot \left(\frac{-1}{\sin h} \right) = e^{h \rightarrow 0} \frac{(1 - \cos h + \sin h)}{\sin h} = e \text{ (LH Rule)}$$

$$\therefore a = e$$

RHL

$$\lim_{h \rightarrow 0} \frac{e^{\frac{1}{h}} + e^{\frac{2}{h}} + e^{\frac{3}{h}}}{e e^{\frac{2}{h}} + b e^{\frac{3}{h}}} = e$$

$$\lim_{h \rightarrow 0} \frac{e^{\frac{1}{h}} + e^{\frac{1}{h}} + 1}{\frac{e}{e^{\frac{1}{h}}} + b} = e$$

(Dividing by $e^{3/h}$)

$$\Rightarrow \frac{1}{b} = e \quad \Rightarrow \quad b = \frac{1}{e}$$

11.(AB) Since intercepts are equal in magnitude but opposite in sign $\Rightarrow \frac{dy}{dx}\bigg|_P = 1$

$$\text{Now } \frac{dy}{dx} = x^2 - 5x + 7 = 1 \Rightarrow x^2 - 5x + 6 = 0 \Rightarrow x = 2 \text{ or } 3$$

12.(BD)
$$h(x) = \frac{\ln(f(x) \cdot g(x))}{\ln a} = \frac{\ln a^{\left\{a^{|x|} \cdot \text{sgn } x\right\} + \left[a^{|x|} \cdot \text{sgn } x\right]}}{\ln a}$$

$$\left(\left\{a^{|x|} \cdot \text{sgn } x\right\} + \left[a^{|x|} \cdot \text{sgn } x\right]\right) = a^{|x|} \cdot \text{sgn } x \quad (\because \{y\} + [y] = y)$$

$$= \begin{cases} a^x & \text{for } x > 0 \\ 0 & \text{for } x = 0 \\ -a^{-x} & \text{for } x < 0 \end{cases} \Rightarrow h(x) \text{ is an odd function}$$

13.(-1)
$$RHD = \lim_{\substack{at \ x=0 \\ h \rightarrow 0}} \frac{h[\sqrt{h} - \sqrt{1+h}] - 0}{h} = -1$$

LHD is not allowed as $x < 0$ is NOT in domain

So, $f'(0) = -1$

14.(-0.67)
$$T_r = \frac{\frac{r(r+1)(2r+1)}{6}}{\frac{r^2(r+1)^2}{4}} = \frac{2r+1}{6} \times \frac{4}{r(r+1)} = \frac{2}{3} \frac{2r+1}{r(r+1)}$$

$$T_r = \frac{2}{3} \left[\frac{1}{r} + \frac{1}{r+1} \right]$$

$$S_n = -T_1 + T_2 - T_3 + T_4 \dots$$

$$= \frac{2}{3} \left[\frac{-1}{1} + \frac{1}{2} - \frac{1}{2} + \frac{1}{3} - \frac{1}{3} + \frac{1}{4} - \frac{1}{4} + \dots \right]$$

$$\therefore \lim_{n \rightarrow \infty} S_n = \frac{-2}{3}$$

15.(0.50)
$$e^{f(x)} = \frac{10+x}{10-x}, \quad x \in (-10, 10)$$

$$f(x) = \log \left(\frac{10+x}{10-x} \right)$$

$$f\left(\frac{200x}{100+x^2}\right) = \log \left[\frac{10 + \frac{200x}{100+x^2}}{10 - \frac{200x}{100+x^2}} \right] = \log \left[\frac{10(10+x)}{10(10-x)} \right]^2 = 2 \log \left(\frac{10+x}{10-x} \right) = 2f(x)$$

$$f(x) = \frac{1}{2} f\left(\frac{200x}{100+x^2}\right) \Rightarrow k = \frac{1}{2} = 0.5$$

16.(0.50)
$$x = -\frac{1}{h}$$

$$\lim_{h \rightarrow 0} \sqrt{\frac{1}{h^2} + \frac{1}{h} + 1} - \frac{a}{h} - b = 0$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{\sqrt{1+h+h^2} - a - bh}{h} = 0$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{1 + \frac{1}{2}(h + h^2) - a - bh}{h} = 0$$

$$\Rightarrow 1 - a = 0 \quad \frac{1}{2} - b = 0$$

$$\Rightarrow a = 1 \quad b = \frac{1}{2}$$

17.(19) Using Lagrange's mean value theorem, for some $c \in (1, 6)$

$$f'(c) = \frac{f(6) - f(1)}{5} = \frac{f(6) + 2}{5} \geq 4.2 \quad \text{or} \quad f(6) + 2 \geq 21 \quad \text{or} \quad f(6) \geq 19$$

18.(-2) Put $\theta = \cos^{-1} \frac{x}{a} \quad (-a < x < a)$

$$\text{Then } 0 < \theta < \pi \text{ and } I = -a \int \frac{(2 + \cos \theta)(1 - \cos \theta)}{1 + \cos \theta} d\theta$$

$$= -a \int \left\{ (1 - \cos \theta) + \frac{1 - \cos \theta}{1 + \cos \theta} \right\} d\theta = -a \left(2 \tan \frac{\theta}{2} - \sin \theta \right) + c = \sqrt{a^2 - x^2} - 2a \sqrt{\frac{a - x}{a + x}} + c$$